Find the Error

Grade 5 · 5.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate (3 + 4) × 2. Student's answer: 11

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied 4 × 2 before working inside the parentheses. Do (3 + 4) first: 7 × 2 = 14.**

Correct answer: 14

Find the Error

Grade 5 · 5.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 20 - (5 + 3). Student's answer: 18

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did 20 - 5 + 3. Work inside the parentheses first: 5 + 3 = 8, then 20 - 8 = 12.**

Correct answer: 12

Find the Error

Grade 5 · 5.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 6 × (2 + 4). Student's answer: 16

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did 6 × 2 + 4. Parentheses come first: 2 + 4 = 6, then 6 × 6 = 36.**

Correct answer: **36**

Find the Error

Grade 5 · 5.OA.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write an expression for: add 6 to 4, then multiply by 3. Student writes: 6 + 4 × 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Parentheses are needed so the sum is multiplied by 3. The correct expression is (4 + 6) × 3.**

Correct answer: **(4 + 6) × 3**

Find the Error

Grade 5 · 5.OA.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write an expression for: subtract 5 from 20, then divide by 3. Student writes: 20 - 5 ÷ 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Without parentheses, only 5 is divided by 3. The correct expression is (20 - 5) ÷ 3.**

Correct answer: **(20 - 5) ÷ 3**

Find the Error

Grade 5 · 5.OA.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write an expression for: twice the sum of 7 and 1. Student writes: 2 + 7 + 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Twice means multiply by 2, and the sum needs parentheses. The correct expression is 2 × (7 + 1).**

Correct answer: **2 × (7 + 1)**

Find the Error

Grade 5 · 5.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The pattern starts at 3 and adds 4 each time: 3, 7, 11, 15, … Find the 5th term. Student's answer: 15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student repeated the 4th term. Add 4 again: 15 + 4 = 19.**

Correct answer: 19

Find the Error

Grade 5 · 5.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Pattern A is 0, 3, 6, 9, … Pattern B is 0, 6, 12, 18, … Compare the matching terms. Student says: B is 3 more than A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student compared the differences between terms. Compare matching terms: 6 ÷ 3 = 2 and 12 ÷ 6 = 2, so each term in B is twice the matching term in A.**

Correct answer: **Each term in B is twice as large as the matching term in A**

Find the Error

Grade 5 · 5.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The pattern starts at 20 and subtracts 2 each time. Find the 5th term. Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 4 from 14 instead of 2. The pattern is 20, 18, 16, 14, 12, so the 5th term is 12.**

Correct answer: 12

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In 2,500, what is the value of the 5? Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The 5 is in the hundreds place, so it is worth 5 hundreds, or 500.**

Correct answer: **500**

Find the Error

Grade 5 · 5.NBT.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In 8,042, what is the value of the 8? Student's answer: 80

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The 8 is in the thousands place, so it is worth 8,000.**

Correct answer: **8,000**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 6 in the hundreds place is how many times as large as the 6 in the tens place? Student's answer: 100 times

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Each place to the left is 10 times the place to its right. Hundreds is 10 times tens.**

Correct answer: **10 times**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is 10^2? Student's answer: 20

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The exponent tells how many 10s to multiply: 10 × 10 = 100.**

Correct answer: **100**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is 3×10^3? Student's answer: 300

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **10^3 means $10 \times 10 \times 10 = 1{,}000$, not 100. So $3 \times 1{,}000 = 3{,}000$.**

Correct answer: **3,000**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is 7.25 × 100? Student's answer: 72.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Multiplying by 100 moves the decimal point two places to the right, not one. The answer is 725.**

Correct answer: **725**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 0.304 in words. Student writes: three hundred four

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The last digit is in the thousandths place, so the words end with thousandths: three hundred four thousandths.**

Correct answer: **three hundred four thousandths**

Find the Error

Grade 5 · 5.NBT.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which is greater: 0.45 or 0.405? Student's answer: 0.405

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Line up the decimals: 0.450 is greater than 0.405 because 0.45 has more tenths and hundredths.**

Correct answer: **0.45**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 2.614 in expanded form. Student writes: $2 + \frac{6}{10} + \frac{4}{100}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The digit 4 is in the thousandths place, so it is $\frac{4}{1}$,000. The student wrote $\frac{4}{100}$, which is a different value. Correct: $2 + \frac{6}{10} + \frac{1}{100} + \frac{4}{1}$,000.**

Correct answer: **$2 + \frac{6}{10} + \frac{1}{100} + \frac{4}{1}$,000**

Find the Error

Grade 5 · 5.NBT.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 3.456 to the nearest hundredth. Student's answer: 3.45

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Look at the thousandths digit, 6. Since 6 is 5 or more, round the hundredths digit up: 3.46.**

Correct answer: **3.46**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 6.282 to the nearest tenth. Student's answer: 6.2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Look at the hundredths digit, 8. Since 8 is 5 or more, round the tenths digit up: 6.3.**

Correct answer: **6.3**

Find the Error

Grade 5 · 5.NBT.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 14.5 to the nearest whole number. Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The tenths digit is 5, so round up. The nearest whole number is 15.**

Correct answer: 15

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply 36 × 24. Student's answer: 764

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The partial products are 144 and 720. Add them: 144 + 720 = 864.**

Correct answer: **864**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply 207 × 5. Student's answer: 135

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied only 7 × 5. Multiply 200 × 5 = 1,000 too. The total is 1,035.**

Correct answer: **1,035**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply 48 × 50. Student's answer: 240

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The 50 has 5 tens, so the answer has an extra zero. 48 × 5 = 240, then × 10 = 2,400.**

Correct answer: 2,400

Find the Error

Grade 5 · 5.NBT.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide 384 ÷ 12. Student's answer: 34

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Check: 12 × 34 = 408, which is too big. 12 × 32 = 384, so the answer is 32.**

Correct answer: **32**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide 936 ÷ 24. Student's answer: 40

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **24 × 40 = 960, which is too big. 24 × 39 = 936, so the answer is 39.**

Correct answer: 39

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide 1,750 ÷ 25. Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **25 × 7 = 175. Since 1,750 is 10 times as large, the answer is 70.**

Correct answer: **70**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add 2.4 + 3.75. Student's answer: 5.15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Line up the decimal points and add 2.40 + 3.75. The answer is 6.15.**

Correct answer: **6.15**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Subtract 8.5 - 3.27. Student's answer: 5.37

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Write 8.50. You need to borrow from the tenths place for the hundredths. The answer is 5.23.**

Correct answer: 5.23

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply 0.6 × 0.4. Student's answer: 2.4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **6 × 4 = 24. The two factors have two decimal places in total, so the answer is 0.24.**

Correct answer: 0.24

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add $\frac{1}{2} + \frac{1}{3}$. Student's answer: $\frac{2}{5}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the numerators and the denominators. Use the common denominator 6: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.**

Correct answer: $\frac{5}{6}$

Find the Error

Grade 5 · 5.NF.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Subtract $\frac{3}{4} - \frac{1}{6}$. Student's answer: $\frac{2}{2}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The student subtracted the denominators too. Use 12: $\frac{9}{12} - \frac{2}{12} = \frac{7}{12}$.

Correct answer: $\frac{7}{12}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add $\frac{2}{3} + \frac{3}{5}$. Student's answer: $\frac{5}{8}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the tops and the bottoms. Use 15: $\frac{10}{15} + \frac{9}{15} = \frac{19}{15}$, which is $1\frac{4}{15}$.**

Correct answer: **$1\frac{4}{15}$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Maria ate $\frac{1}{3}$ of a pizza. Leo ate $\frac{1}{4}$ of the same pizza. How much did they eat in all? Student's answer: $\frac{2}{7}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The student added the denominators. Use 12: $\frac{4}{12} + \frac{3}{12} = \frac{7}{12}$.

Correct answer: $\frac{7}{12}$ of the pizza

Find the Error

Grade 5 · 5.NF.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bottle has $\frac{3}{4}$ liter of juice. Kai pours out $\frac{1}{2}$ liter. How much is left? Student's answer: $\frac{1}{2}$ liter

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{1}{2}$ liter is what was poured out, not what is left. $\frac{3}{4} - \frac{1}{2} = \frac{3}{4} - \frac{2}{4} = \frac{1}{4}$ liter.

Correct answer: $\frac{1}{4}$ liter

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Without finding the exact sum, is $\frac{5}{9} + \frac{7}{10}$ more or less than 1? Student says: less than 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$\frac{5}{9}$ is a little more than $\frac{1}{2}$, and $\frac{7}{10}$ is more than $\frac{1}{2}$. Together they are more than 1, so the sum is more than 1.**

Correct answer: **More than 1**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 3 ÷ 4 as a fraction. Student writes: $\frac{4}{3}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The number being divided goes on top. 3 ÷ 4 = $\frac{3}{4}$.

Correct answer: $\frac{3}{4}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

5 sandwiches are shared equally by 2 people. How much does each person get? Student's answer: $\frac{2}{5}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student flipped the numbers. $5 \div 2 = \frac{5}{2} = 2\ \frac{1}{2}$ sandwiches each.**

Correct answer: **$2\ \frac{1}{2}$ sandwiches**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write $\frac{7}{3}$ as a mixed number. Student's answer: $3\ \frac{1}{3}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: 7 ÷ 3 = 2 remainder 1. So $\frac{7}{3} = 2\ \frac{1}{3}$.

Correct answer: $2\ \frac{1}{3}$

Find the Error

Grade 5 · 5.NF.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $\frac{2}{3} \times \frac{3}{5}$. Student's answer: $\frac{6}{8}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The student added the denominators. Multiply: 2 × 3 = 6 and 3 × 5 = 15, so $\frac{6}{15}$. Simplify to $\frac{2}{5}$.

Correct answer: $\frac{2}{5}$

Find the Error

Grade 5 · 5.NF.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle is $\frac{3}{4}$ foot wide and $\frac{2}{3}$ foot long. What is its area? Student's answer: $\frac{5}{7}$ square foot

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Area is length times width, so multiply: $\frac{3}{4} \times \frac{2}{3} = \frac{6}{12} = \frac{1}{2}$ square foot.**

Correct answer: **$\frac{1}{2}$ square foot**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is $\frac{3}{4}$ of 12? Student's answer: 36

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied 3 × 12 and kept the 4 as the denominator. Divide 12 by 4 first, then multiply by 3: 9.**

Correct answer: 9

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Without multiplying, is $6 \times \frac{2}{3}$ larger or smaller than 6? Student says: larger

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{2}{3}$ **is less than 1, so multiplying by it makes the number smaller. The answer is 4, which is smaller than 6.**

Correct answer: **Smaller**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Without multiplying, is $5 \times \frac{3}{2}$ larger or smaller than 5? Student says: smaller

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{3}{2}$ is greater than 1, so multiplying by it makes the number larger. The answer is 7 $\frac{1}{2}$.

Correct answer: Larger

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Without multiplying, is $9 \times \frac{5}{5}$ equal to 9, larger than 9, or smaller than 9? Student says: larger

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{5}{5}$ equals 1, so the product stays the same. $9 \times \frac{5}{5} = 9$.

Correct answer: Equal to 9

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rope is 12 feet long. Ben cuts off $\frac{1}{4}$ of it. Then he cuts off $\frac{1}{3}$ of the rest. How long is the piece left? Student's answer: 9 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student stopped after the first cut. After the first cut, 9 feet is left. Then $\frac{1}{3}$ of 9 is 3 feet, so 6 feet is left.**

Correct answer: **6 feet**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A hiker walks $\frac{2}{5}$ mile, then $\frac{3}{10}$ mile, then $\frac{1}{2}$ mile. How far does she walk in all? Student's answer: $2\ \frac{1}{5}$ miles

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The fractions in tenths are $\frac{4}{10}$, $\frac{3}{10}$, and $\frac{5}{10}$. They add to $\frac{12}{10}$, which is $1\ \frac{2}{10}$ or $1\ \frac{1}{5}$ miles.

Correct answer: $1\ \frac{1}{5}$ miles

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A basket has 24 apples. $\frac{1}{4}$ of them are red. Half of the remaining apples are green. How many green apples are there? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student took half of all 24 apples. First find the remaining apples: 24 - 6 = 18. Half of 18 is 9.**

Correct answer: 9

Find the Error

Grade 5 · 5.NF.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide $\frac{1}{3} \div 2$. Student's answer: $\frac{2}{3}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied by 2. Dividing $\frac{1}{3}$ into 2 equal parts gives $\frac{1}{6}$.**

Correct answer: $\frac{1}{6}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide $4 \div \frac{1}{5}$. Student's answer: $\frac{4}{5}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **There are 5 fifths in each whole, so there are 4 × 5 = 20 fifths in 4 wholes.**

Correct answer: 20

Find the Error

Grade 5 · 5.NF.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide $\frac{1}{2} \div 4$. Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided the other way. Split $\frac{1}{2}$ into 4 equal parts: $\frac{1}{8}$.**

Correct answer: $\frac{1}{8}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Convert 3 feet to inches. Student's answer: 30 inches

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **There are 12 inches in 1 foot. So 3 × 12 = 36 inches.**

Correct answer: **36 inches**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Convert 4 kilograms to grams. Student's answer: 400 grams

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **There are 1,000 grams in 1 kilogram. So 4 × 1,000 = 4,000 grams.**

Correct answer: **4,000 grams**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Convert 2.5 liters to milliliters. Student's answer: 250 milliliters

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **There are 1,000 milliliters in 1 liter. So 2.5 × 1,000 = 2,500 milliliters.**

Correct answer: **2,500 milliliters**

Find the Error

Grade 5 · 5.MD.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line plot shows these lengths in inches: $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, 1, $\frac{1}{4}$. What is the total of all the lengths?

Student's answer: 3 inches

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: The total is $\frac{1}{4} + \frac{1}{2} + \frac{1}{2} + \frac{3}{4} + 1 + \frac{1}{4} = 3\ \frac{1}{4}$ inches. The student left out the fraction.

Correct answer: $3\ \frac{1}{4}$ inches

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line plot shows $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, and 1 $\frac{1}{4}$ inches. How many measurements are less than 1 inch?

Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The measurements $\frac{1}{2}$, $\frac{1}{2}$, and $\frac{3}{4}$ are all less than 1 inch, so the count is 3.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Four bean sprouts grew these amounts in inches: $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{4}$, $\frac{3}{8}$. What is the total growth?

Student's answer: $\frac{1}{2}$ inch

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Use eighths: $\frac{1}{8} + \frac{2}{8} + \frac{2}{8} + \frac{3}{8} = \frac{8}{8} = 1$ inch.**

Correct answer: **1 inch**

Find the Error

Grade 5 · 5.MD.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A cube has sides that are each 1 unit long. What is its volume? Student's answer: 1 square unit

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Volume is measured in cubic units because it counts cubes. The answer is 1 cubic unit.**

Correct answer: **1 cubic unit**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box has 3 layers of 8 unit cubes each. What is its volume? Student's answer: 11 cubic units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 3 + 8. Multiply the layers by the cubes in each layer: 3 × 8 = 24 cubic units.**

Correct answer: **24 cubic units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which unit measures volume: square centimeter or cubic centimeter? Student's answer: square centimeter

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Square units measure area. Volume uses cubic units, such as cubic centimeters.**

Correct answer: **Cubic centimeter**

Find the Error

Grade 5 · 5.MD.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A solid has 4 layers, and each layer is 3 by 2 unit cubes. How many cubes are in the solid?

Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Each layer has 3 × 2 = 6 cubes. With 4 layers, the total is 4 × 6 = 24 cubes.**

Correct answer: **24 cubes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A solid has 2 layers, and each layer is a 3 by 3 square of unit cubes. How many cubes are in the solid? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted only one layer. Each layer has 9 cubes, so 2 layers have 18 cubes.**

Correct answer: **18 cubes**

Find the Error

Grade 5 · 5.MD.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangular prism is built with unit cubes. It is 5 cubes long, 2 cubes wide, and 3 cubes tall. How many cubes are in it? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied only two dimensions. 5 × 2 × 3 = 30 cubes.**

Correct answer: **30 cubes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box is 4 cm long, 3 cm wide, and 2 cm tall. What is its volume? Student's answer: 9 cubic cm

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the sides. Multiply: 4 × 3 × 2 = 24 cubic cm.**

Correct answer: **24 cubic cm**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box is 6 units long, 5 units wide, and 2 units tall. What is its volume? Student's answer: 22 cubic units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 6 + 5 and then multiplied by 2. Multiply all three: 6 × 5 × 2 = 60 cubic units.**

Correct answer: **60 cubic units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box is 1.5 units long, 2 units wide, and 4 units tall. What is its volume? Student's answer: 7.5 cubic units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the sides or made a multiplication mistake. 1.5 × 2 = 3, and 3 × 4 = 12 cubic units.**

Correct answer: **12 cubic units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the x-coordinate of the point (3, 5)? Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The first number in an ordered pair is the x-coordinate. For (3, 5), the x-coordinate is 3.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Plot the point (4, 2). The student plotted the point at (2, 4). Is this correct? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The x-coordinate comes first. For (4, 2), go 4 units right and 2 units up, not the other way.**

Correct answer: **No. The point is at (4, 2)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the ordered pair for the origin? Student's answer: (1, 1)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The origin is where the axes meet, at zero on both axes. Its ordered pair is (0, 0).**

Correct answer: **(0, 0)**

Find the Error

Grade 5 · 5.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A graph shows hours studied on the x-axis and score on the y-axis. Which point shows 2 hours and a score of 60? Student's answer: (60, 2)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Hours are on the x-axis, so they come first. The point is (2, 60).**

Correct answer: **(2, 60)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through (0, 0) and (4, 8). Is the point (2, 4) on the line? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The line goes up 2 for every 1 across. (2, 4) fits: 2 is half of 4, and 4 is half of 8. It is on the line.**

Correct answer: **Yes**

Find the Error

Grade 5 · 5.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

3 pencils cost 15 cents. Write the point that represents this, with pencils on the x-axis and cents on the y-axis. Student's answer: (15, 3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Pencils are on the x-axis, so the point is (3, 15).**

Correct answer: **(3, 15)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is every square a rectangle? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A square has four right angles and two pairs of parallel sides, so it meets the rules of a rectangle. Every square is a rectangle.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is every rectangle a square? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A square needs all four sides equal. A 4 by 2 rectangle has four right angles but is not a square.**

Correct answer: **No**

Find the Error

Grade 5 · 5.G.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is every rhombus a square? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A rhombus has four equal sides, but it does not need right angles. A square needs right angles too.**

Correct answer: **No**

Find the Error

Grade 5 · 5.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has three sides that are each 3 cm long. How can you classify it by its sides? Student's answer: scalene

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A scalene triangle has no equal sides. All three sides are equal, so it is equilateral.**

Correct answer: **Equilateral**

Find the Error

Grade 5 · 5.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has angles of 30°, 60°, and 90°. How can you classify it by its angles? Student's answer: acute

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **One angle is exactly 90°, so the triangle is a right triangle. An acute triangle has all angles less than 90°.**

Correct answer: **Right triangle**

Find the Error

Grade 5 · 5.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A quadrilateral has four right angles and two pairs of parallel sides, but the sides are not all equal. What is it? Student's answer: rhombus

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A rhombus has four equal sides, but this shape does not. Four right angles and parallel sides make it a rectangle.**

Correct answer: **Rectangle**